

KiCad CLI & CI/CD Integration

Headless KiCad — Gerber gen, ERC/DRC, BOM, 3D — automated

Software-style continuous integration for hardware

Audience: PCB Engineers + DevOps-curious

kiCad-cli • KiBot • GitHub Actions • GitLab CI • Best practices

Why CI for KiCad

Auto Gerber

Every commit produces
fab-ready files.

ERC/DRC gates

Catch issues at PR
before merge.

Always-current BOM

No manual export drift.

Visual diffs

kicad-diff in PR comments
for schematic + PCB.

Release packages

Tag commits → fab fileset
auto-attached.

Quality gate

Block merges with errors
— no bad PCBs ship.

kicad-cli Overview

- ▶ kicad-cli: built-in CLI tool, shipped with KiCad 8+
- ▶ Runs HEADLESS — no GUI required
- ▶ Subcommands: pcb, sch, fp (footprint), sym (symbol), python
- ▶ Format: 'kicad-cli <command> <subcommand> [options] <file>'
- ▶ Common: 'kicad-cli pcb export gerbers', 'kicad-cli sch export pdf'
- ▶ Output: standardised, parseable
- ▶ Used by: KiBot, custom CI scripts, internal build pipelines

Common kicad-cli Commands

Schematic exports

```
kicad-cli sch export pdf board.kicad_sch --output schematic.pdf  
kicad-cli sch export bom board.kicad_sch --output bom.csv  
kicad-cli sch export netlist board.kicad_sch --format kicadsexpr
```

PCB exports

```
kicad-cli pcb export gerbers board.kicad_pcb --output gerber/  
kicad-cli pcb export drill board.kicad_pcb --output gerber/  
kicad-cli pcb export pos board.kicad_pcb --output pos.csv --format csv  
kicad-cli pcb export step board.kicad_pcb --output board.step  
kicad-cli pcb export svg board.kicad_pcb --output board.svg
```

DRC + ERC checks

```
kicad-cli pcb drc board.kicad_pcb --output drc.json --format json  
kicad-cli sch erc board.kicad_sch --output erc.json --format json
```

KiBot — High-Level Automation

- ▶ KiBot: github.com/INTI-CMNB/KiBot
- ▶ YAML-configured automation tool — wraps kicad-cli + custom scripts
- ▶ Outputs: Gerber, BOM, drill, position, 3D model, PDF schematic, ERC/DRC reports
- ▶ Single config file (.kibot.yaml) drives all outputs
- ▶ Pre-flight: runs ERC + DRC before any export
- ▶ Integration: GitHub Actions, GitLab CI, Drone, Jenkins
- ▶ Docker images available: [setsoft/kicad_auto](https://setsoft.com/kicad_auto)
- ▶ De-facto standard for KiCad CI in 2026

Sample KiBot Config

```
# .kibot.yaml
kibot:
  version: 1

preflight:
  run_erc: true
  run_drc: true
  update_xml: true
  check_zone_fills: true

outputs:
  - name: gerbers
    type: gerber
    dir: gerber/
  - name: drill
    type: excellon
    dir: gerber/
  - name: bom
    type: bom
    dir: bom/
    options:
      format: HTML
  - name: position
    type: position
    dir: gerber/
  - name: 3d
    type: render_3d
    dir: 3d/
  - name: schematic
    type: pdf_sch_print
```

GitHub Actions Workflow

```
# .github/workflows/build.yml
name: Build PCB Outputs
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

jobs:
  build:
    runs-on: ubuntu-latest
    container:
      image: setsoft/kicad_auto:dev_k8
    steps:
      - uses: actions/checkout@v4

      - name: Run KiBot
        run: kikit -c .kikit.yaml -d output

      - name: Upload Gerbers
        uses: actions/upload-artifact@v4
        with:
          name: gerbers
          path: output/gerber/
```

Pull-Request Workflow

- ▶ Designer commits change → opens PR
- ▶ GitHub Actions runs: ERC + DRC + Gerber + schematic PDF
- ▶ Failed ERC or DRC → blocks merge automatically
- ▶ kicad-diff plugin: visual schematic + PCB diff in PR comment
- ▶ Reviewer sees: changed-file list + visual rendering + artifacts
- ▶ Approves → merge into main
- ▶ On main: same build runs, attaches artifacts to commit
- ▶ On tag (v1.0): build runs, creates GitHub Release with fab fileset

Release Pipeline

```
# .github/workflows/release.yml
name: Release Build
on:
  push:
    tags: ['v*']

jobs:
  release:
    runs-on: ubuntu-latest
    container:
      image: setsoft/kicad_auto:dev_k8
    steps:
      - uses: actions/checkout@v4
      - name: Generate fab package
        run: kikit -c .kikit.yaml -d output

      - name: Zip artifacts
        run: cd output && zip -r fab-package.zip .

      - name: Create GitHub Release
        uses: softprops/action-gh-release@v2
        with:
          files: output/fab-package.zip
          generate_release_notes: true
```

Self-Hosted Alternatives

GitLab CI

Self-host runner
Same YAML pattern

Drone CI

Lightweight, Docker-native

Buildkite

Mid-size organizations
Good UI

Jenkins

Old-school, powerful
Learning curve

Local Makefile

Simplest — no CI infra
For solo developers

GitHub free tier

2k min/mo public repos
Usually enough for HW projects

Common CI Issues

- ▶ KiCad version mismatch → pin Docker image tag (e.g., 'kicad_auto:dev_k8')
- ▶ Custom library paths break in CI → use relative paths
- ▶ Large 3D models slow runner → use LFS or skip 3D in CI
- ▶ Plugin dependencies → install in Dockerfile or per-step
- ▶ Stale fp-info-cache → rebuild explicitly in script
- ▶ Symbols renamed across rev → update KiBot config
- ▶ GitHub free tier minutes exhausted → self-host runner on Raspberry Pi
- ▶ Errors only in artifacts → surface in PR comment

Beyond Gerber CI

- ▶ Firmware CI: compile, unit-test FW for the same PCB project
- ▶ Hardware-in-loop: connect dev-board to runner — live test
- ▶ 3D STEP export: for mech engineers (auto-update when PCB changes)
- ▶ Cost analysis: OctoPart API computes BOM cost on every commit
- ▶ Component lifecycle: flag EOL parts before redesign
- ▶ Sim runs: batch SPICE on every commit
- ▶ Test report: HTML dashboard linked from CI status

Key Takeaways

Key takeaways from this lecture

kicad-cli is built in

KiCad 8+ ships with it.

KiBot = best wrapper

YAML config drives everything.

GitHub Actions = free CI

Public repos, 2k min/mo.

ERC/DRC = quality gate

Block merges with errors.

Release pipeline

Tag → fab package auto-attached.

Docker for reproducibility

Pin KiCad version explicitly.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ KiBot — github.com/INTI-CMNB/KiBot
- ▶ kicad-cli reference (docs.kicad.org)
- ▶ GitHub Actions documentation
- ▶ GitLab CI documentation
- ▶ Robert Feranec / Phil's Lab — CI for KiCad videos
- ▶ [setsoft/kicad_auto](#) Docker image
- ▶ KiCanvas — embed KiCad rendering in PR comments
- ▶ INTI-CMNB workflows on GitHub

Thank you • Build something this week.